

Comprehensive Pedagogical Evaluation of a Computational Tool for Bioprocess Modeling, Optimization, and Control

CABBIO BR03 - Bioprocess Engineering Education Research

Version 2.0 - 2026

Research Framework

Purpose: This survey instrument evaluates the pedagogical effectiveness of an interactive computational tool designed for teaching bioprocess modeling, parameter estimation, optimization, and control strategies in chemical, agroindustrial, and bioprocess engineering curricula.

Theoretical Framework: This evaluation is grounded in Cognitive Load Theory (Sweller, 1988), Constructivist Learning Theory (Piaget, Vygotsky), and the ABET Engineering Criteria for student outcomes assessment.

Metrics Generated:

- **Usability Index (UI):** Sections B + C
- **Cognitive Load Reduction Index (CLRI):** Sections D + E
- **Learning Effectiveness Index (LEI):** Sections F + G + H + I
- **Industry Readiness Index (IRI):** Section J
- **Overall Pedagogical Effectiveness Score (OPES):** Weighted composite
- **Net Promoter Score (NPS):** Section K, Item 45

Instructions: For Likert-scale items, mark with an 'X' where:

1 = Strongly Disagree — 2 = Disagree — 3 = Neutral — 4 = Agree — 5 = Strongly Agree

Estimated completion time: 20-25 minutes

1 A. Participant Demographics and Prior Knowledge Profile

This section establishes baseline characteristics for stratified analysis.

A1. Full Name (Optional):	A2. Email (Optional):

A3. Academic Program:

- ☐ Chemical Engineering ☐ Bioprocess Engineering ☐ Agroindustrial Engineering
☐ Biotechnology ☐ Food Engineering ☐ Environmental Engineering
☐ Other: _____

A4. Current Academic Level:

- ☐ Undergraduate (1st-2nd year) ☐ Undergraduate (3rd-4th year) ☐ Final year/Thesis
☐ Master's Student ☐ PhD Student ☐ Faculty/Instructor ☐ Industry Professional

A5. Institution Name: _____ **Country:** _____

A6. Age Range:

- ☐ 18-22 ☐ 23-27 ☐ 28-35 ☐ 36-45 ☐ ≥45

A7. Gender:

☐ Male ☐ Female ☐ Non-binary ☐ Prefer not to say

Prior Knowledge Self-Assessment (1 = None, 5 = Expert):

Knowledge Area	1	2	3	4	5
A8. Differential Equations (ODEs/PDEs)	☐	☐	☐	☐	☐
A9. Bioprocess Fundamentals (microbial growth, kinetics)	☐	☐	☐	☐	☐
A10. Process Modeling and Simulation	☐	☐	☐	☐	☐
A11. Process Control Theory (PID, feedback loops)	☐	☐	☐	☐	☐
A12. Advanced Control (MPC, Fuzzy, Adaptive)	☐	☐	☐	☐	☐
A13. Parameter Estimation / Data Fitting	☐	☐	☐	☐	☐
A14. Python Programming	☐	☐	☐	☐	☐
A15. Other Simulation Tools (MATLAB, Aspen, etc.)	☐	☐	☐	☐	☐

A16. Time spent using this tool:

☐ < 1 hour ☐ 1-3 hours ☐ 3-6 hours ☐ 6-10 hours ☐ > 10 hours

A17. Context of tool usage:

☐ Classroom instruction ☐ Laboratory session ☐ Homework/Assignment
☐ Self-directed learning ☐ Research project ☐ Other: _____

2 B. System Usability and Technical Performance

*Evaluates extrinsic cognitive load and technical reliability. Contributes to **Usability Index (UI)**.*

Statement	1	2	3	4	5
B1. The installation/setup process was straightforward.	☐	☐	☐	☐	☐
B2. The main dashboard layout is logically organized.	☐	☐	☐	☐	☐
B3. Navigation between modules (Batch, Continuous, Control, Estimation) is intuitive.	☐	☐	☐	☐	☐
B4. The tool responds quickly without noticeable lag.	☐	☐	☐	☐	☐
B5. The application runs stably without crashes or unexpected errors.	☐	☐	☐	☐	☐
B6. Error messages, when they occur, are informative and helpful.	☐	☐	☐	☐	☐
B7. I could use the tool effectively without external technical support.	☐	☐	☐	☐	☐

3 C. Visual Design and Information Presentation

*Assesses graphical clarity and information architecture. Contributes to **UI** and **CLRI**.*

Statement	1	2	3	4	5
C1. Graphical plots of state variables (biomass X , substrate S , product P) are clear and readable.	☐	☐	☐	☐	☐
C2. Axis labels, units, and legends are appropriately displayed.	☐	☐	☐	☐	☐
C3. Color coding helps distinguish between different variables and scenarios.	☐	☐	☐	☐	☐
C4. Parameter input fields are clearly labeled with appropriate units.	☐	☐	☐	☐	☐
C5. It is easy to modify kinetic parameters (e.g., μ_{max} , K_s , K_i).	☐	☐	☐	☐	☐
C6. It is easy to modify control parameters (e.g., PID gains K_p , K_i , K_d).	☐	☐	☐	☐	☐
C7. Real-time simulation updates help visualize dynamic behavior effectively.	☐	☐	☐	☐	☐
C8. The mathematical equations displayed alongside simulations aid understanding.	☐	☐	☐	☐	☐

4 D. Cognitive Load Assessment

*Measures mental effort during tool usage. Contributes to **CLRI**. Based on NASA-TLX adapted items.*

Statement	1	2	3	4	5
D1. Using this tool required less mental effort than studying the same concepts from textbooks alone.	☐	☐	☐	☐	☐
D2. The tool reduced the complexity of understanding bioprocess dynamics.	☐	☐	☐	☐	☐
D3. I did not feel overwhelmed by the amount of information presented.	☐	☐	☐	☐	☐
D4. The progressive complexity (from simple to advanced modules) was appropriate.	☐	☐	☐	☐	☐
D5. I could focus on learning concepts rather than struggling with the interface.	☐	☐	☐	☐	☐
D6. The tool's feedback (graphs, numerical outputs) helped me self-assess my understanding.	☐	☐	☐	☐	☐

5 E. Learning Effectiveness: Theoretical Foundations

*Evaluates understanding of underlying principles. Contributes to **LEI**.*

Statement	1	2	3	4	5
E1. The tool helped me understand how ODEs represent mass balances in bioreactors.	☐	☐	☐	☐	☐
E2. I can better explain the mathematical formulation: $\frac{dX}{dt} = \mu X - DX$	☐	☐	☐	☐	☐
E3. I understand the physical meaning of yield coefficients ($Y_{X/S}$, $Y_{P/S}$, $Y_{P/X}$).	☐	☐	☐	☐	☐
E4. The tool clarified how different kinetic models (Monod, Haldane, Contois) behave.	☐	☐	☐	☐	☐
E5. I understand the role of μ_{max} , K_s , and K_i in determining growth dynamics.	☐	☐	☐	☐	☐
E6. I can now represent substrate inhibition mathematically using Haldane kinetics.	☐	☐	☐	☐	☐
E7. The simulator clarified the difference between growth-associated and non-growth-associated product formation.	☐	☐	☐	☐	☐

6 F. Learning Effectiveness: Bioreactor Operation Modes

*Assesses understanding of operational strategies. Contributes to **LEI**.*

Statement	1	2	3	4	5
F1. I clearly understand the dynamic behavior of Batch fermentation.	☐	☐	☐	☐	☐
F2. I can differentiate between Fed-Batch feeding strategies (constant, exponential, optimal).	☐	☐	☐	☐	☐
F3. The Continuous (CSTR) module helped me understand steady-state operation and washout conditions.	☐	☐	☐	☐	☐
F4. I understand how dilution rate (D) affects productivity and cell concentration.	☐	☐	☐	☐	☐
F5. The tool helped me visualize the trade-offs between different operation modes.	☐	☐	☐	☐	☐
F6. I can now select an appropriate bioreactor operation mode for a given production objective.	☐	☐	☐	☐	☐

7 G. Learning Effectiveness: Parameter Estimation

*Evaluates data-driven modeling skills. Contributes to **LEI**.*

Statement	1	2	3	4	5
G1. The Parameter Estimation module improved my understanding of least-squares fitting.	☐	☐	☐	☐	☐
G2. I understand the concept of objective function minimization in parameter estimation.	☐	☐	☐	☐	☐
G3. The tool helped me understand parameter identifiability and sensitivity.	☐	☐	☐	☐	☐
G4. I can interpret confidence intervals and parameter uncertainty from estimation results.	☐	☐	☐	☐	☐
G5. The visualization of fitted vs. experimental data clarified model validation concepts.	☐	☐	☐	☐	☐
G6. I understand how an Extended Kalman Filter (EKF) estimates states from noisy measurements.	☐	☐	☐	☐	☐
G7. The tool illustrated the importance of model-experiment reconciliation.	☐	☐	☐	☐	☐

8 H. Learning Effectiveness: Regulatory Control (PID)

*Assesses classical control understanding. Contributes to **LEI**.*

Statement	1	2	3	4	5
H1. I understand the individual effects of K_p , K_i , and K_d on closed-loop response.	☐	☐	☐	☐	☐
H2. The simulator helped me visualize overshoot, settling time, and steady-state error.	☐	☐	☐	☐	☐
H3. I can tune a PID controller to meet performance specifications using this tool.	☐	☐	☐	☐	☐
H4. I understand feedback control applied to dissolved oxygen (DO) or temperature regulation.	☐	☐	☐	☐	☐
H5. The cascade control module clarified master-slave controller configurations.	☐	☐	☐	☐	☐
H6. I understand the limitations of PID control for nonlinear bioprocesses.	☐	☐	☐	☐	☐

9 I. Learning Effectiveness: Advanced Control and Optimization

*Evaluates understanding of modern control techniques. Contributes to **LEI**.*

Statement	1	2	3	4	5
I1. The NMPC module helped me understand model-based predictive control.	☐	☐	☐	☐	☐
I2. I understand the role of prediction horizon (N_p) and control horizon (N_c) in MPC.	☐	☐	☐	☐	☐
I3. The tool clarified how constraints are handled in MPC optimization.	☐	☐	☐	☐	☐
I4. I can compare the performance advantages of MPC over PID for bioprocesses.	☐	☐	☐	☐	☐
I5. The Fuzzy Logic control module helped me understand rule-based control systems.	☐	☐	☐	☐	☐
I6. I understand Real-Time Optimization (RTO) and its role in economic process operation.	☐	☐	☐	☐	☐
I7. The tool demonstrated the hierarchy: RTO $\rightarrow$ MPC $\rightarrow$ Regulatory Control.	☐	☐	☐	☐	☐
I8. I can formulate a simple optimization problem for a fed-batch process.	☐	☐	☐	☐	☐

10 J. Industry Readiness and Professional Competencies

*Assesses preparation for professional practice. Contributes to **IRI**.*

Statement	1	2	3	4	5
J1. This tool bridges the gap between theoretical equations and real bioreactor operation.	☐	☐	☐	☐	☐
J2. I feel more confident discussing bioprocess control with industry professionals.	☐	☐	☐	☐	☐
J3. The skills practiced here are relevant to pharmaceutical/biotechnology industries.	☐	☐	☐	☐	☐
J4. I am better prepared to design control strategies for industrial bioreactors.	☐	☐	☐	☐	☐
J5. The tool enhanced my problem-solving abilities in bioprocess engineering.	☐	☐	☐	☐	☐
J6. Using this simulator is a valuable complement to laboratory experiments.	☐	☐	☐	☐	☐
J7. I would use similar computational tools in my future professional work.	☐	☐	☐	☐	☐
J8. This experience improved my ability to interpret process data and make decisions.	☐	☐	☐	☐	☐

11 K. Overall Satisfaction and Recommendation

Global assessment for **OPES** and **NPS** calculation.

Statement	1	2	3	4	5
K1. Overall, this tool met my learning expectations.	☐	☐	☐	☐	☐
K2. The time invested using this tool was worthwhile for my education.	☐	☐	☐	☐	☐
K3. This tool is superior to other simulation software I have used for similar topics.	☐	☐	☐	☐	☐
K4. I would recommend this tool to fellow students in bioprocess engineering.	☐	☐	☐	☐	☐

K5. Net Promoter Score (NPS): On a scale of 0-10, how likely are you to recommend this tool to a colleague or classmate?

0	1	2	3	4	5	6	7	8	9	10
☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐
Not at all likely			Neutral				Extremely likely			

12 L. Pre-Post Knowledge Assessment (Optional - For Instructors)

Direct knowledge measurement for learning gains analysis. Use identical items before and after tool usage.

Instructions: Rate your confidence in answering the following questions correctly (1 = Cannot answer, 5 = Very confident).

Concept Assessment	1	2	3	4	5
L1. Write the ODE for biomass balance in a continuous bioreactor.	☐	☐	☐	☐	☐
L2. Explain why Haldane kinetics shows a maximum in the μ vs. S curve.	☐	☐	☐	☐	☐
L3. Calculate the critical dilution rate for washout in a chemostat.	☐	☐	☐	☐	☐
L4. Describe the effect of increasing K_i in a PID controller.	☐	☐	☐	☐	☐
L5. Explain the difference between prediction and control horizons in MPC.	☐	☐	☐	☐	☐
L6. Describe how an EKF estimates unmeasured states in a bioprocess.	☐	☐	☐	☐	☐

13 M. Qualitative Feedback and Open-Ended Questions

Rich qualitative data for thematic analysis.

M1. Which specific concept(s) became clearer after using this tool?

M2. Which module did you find most valuable for your learning? Why?

- ☐ Batch Modeling
 ☐ Fed-Batch
 ☐ Continuous
 ☐ PID Control
☐ Cascade Control
 ☐ NMPC
 ☐ Fuzzy Control
 ☐ RTO
 ☐ Parameter Estimation

Explanation:

M3. Which module requires additional theoretical explanations or documentation within the interface?

- ☐ Batch Modeling
 ☐ Fed-Batch
 ☐ Continuous
 ☐ PID Control
☐ Cascade Control
 ☐ NMPC
 ☐ Fuzzy Control
 ☐ RTO
 ☐ Parameter Estimation

Specific suggestions:

M4. What additional bioprocesses or case studies would you like to see included?
(e.g., wastewater treatment, enzyme production, cell culture, anaerobic digestion)

M5. What features would improve the tool's educational value?

- ☐ Video tutorials
 ☐ Guided exercises
 ☐ More theory explanations
☐ Quiz/assessment integration
 ☐ Multi-language support
 ☐ Mobile version
☐ Collaborative features
 ☐ Export capabilities
 ☐ Other: _____

Additional comments:

M6. Please describe any difficulties or frustrations you experienced while using the tool:

M7. General comments, suggestions, or any other feedback:

Statistical Analysis Framework

For Researchers: The following metrics can be computed from this survey:

Metric	Calculation	Sections
Usability Index (UI)	Mean of B1-B7, C1-C8	B, C
Cognitive Load Reduction Index (CLRI)	Mean of D1-D6	D
Learning Effectiveness Index (LEI)	Mean of E1-E7, F1-F6, G1-G7, H1-H6, I1-I8	E, F, G, H, I
Industry Readiness Index (IRI)	Mean of J1-J8	J
Overall Pedagogical Effectiveness Score	$0.15 \times UI + 0.15 \times CLRI + 0.50 \times LEI + 0.20 \times IRI$	Composite
Net Promoter Score (NPS)	%Promoters (9-10) - %Detractors (0-6)	K5
Normalized Learning Gain	$(Post - Pre) / (5 - Pre)$ for Section L items	L

Recommended Statistical Tests:

- Cronbach's α for internal consistency of each construct
- Paired t-test or Wilcoxon signed-rank for pre-post comparisons (Section L)
- ANOVA/Kruskal-Wallis for comparing groups (by program, prior knowledge level)
- Spearman correlation between prior knowledge (A8-A15) and learning outcomes
- Factor analysis to validate construct structure
- Thematic analysis for open-ended responses (Section M)

Thank you for your participation!

Your feedback is essential for improving bioprocess engineering education.

For questions about this survey, contact: [instructor email]